Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*AktiMeter: Is a Camera based technology that allows detecting human activities, intentions, driver states and interactions in vehicle cabin (driving and non-driving) and in assembly*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (x), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (x), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electronics		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	*Optical Measuring Instruments*	*R&D / Engineering Services*	*Testing / Inspection / Analysis Techniques*
Concept of Proposed Technology	AktiMeter is a *Real Time and Edge Image Processing with Visual Language Models to support humans in vehicles or assembly. It is used for benchmarks, labelling user activities, AI data collection on the road, prototyping, user testing, and final scalable product integration*			
Rationale for Joint R&D Project	*AktiMeter is a unique measuring product based on AI, optimized for in vehicle use cases and for assembly use cases. The R&D shall extend the use cases, enable new products, better user studies and better user experience, market insights and user centred prototyping, etc.*			
Key Words	1. *Camera* 2. *Visual Language Models* 3. *InCabin Sensing* 4. *Edge AI* 5. *User Testing*			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	*Level (8) AktiMeter development status is 8. Can be used in TRL 4-9 projects*	**End of the Project**	Level (9)
R&D Objectives	*Chatbot with Eyes in the car / truck, Personalized user experience in the car, AI guided Assembly and documentation.*			
R&D Target Specifications	Chatbot with Eyes: *Detect complex activities, predict activities, etc. and base chatbot assistance on this information*			
R&D Plan	*Define use cases for chatbot with eyes* *Develop Proof of Concept Demonstrator* *User Testing* *Edge AI optimization* *Validation*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Setting up measuring system, setting up PoC, performing user studies* Providing driving simulator and Mercedes EQS Drive Pilot Level 3 road car
Expected R&R of Korean SME Partner	*InCabin use case* *Assembly use case*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Expected outcome is a product prototype (e.g. chatbot with eyes) or a product (assembly assist)*

6. References

(1) https://www.iosb.fraunhofer.de/en/projects-and-products/aktimeter.html

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization